Observational Study

Medicine

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The association between asthma and ADHD in children and adolescents

An observational study and a meta-analysis of Mendelian randomization studies

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Abstract

Observational epidemiological analysis and a meta-analysis of Mendelian randomization (MR) were used to investigate the association between attention-deficit/hyperactivity disorder (ADHD) and asthma in children and adolescents. Based on data from the 1999 to 2004 National Health and Nutrition Examination Survey (NHANES), an analysis was conducted to evaluate the association between ADHD and asthma in children and adolescents aged between 4 and 19 years. ADHD and asthma were defined based on self- or parent/guardian proxy-reported physician/health-professional diagnoses from the NHANES Medical Conditions Questionnaire. After analyzing ADHD genome-wide association study data from the Psychiatric Genomics Consortium as instrumental variables, we performed two-sample MR analyses based on genome-wide association study data from 5 studies to examine the causal relationship between genetically predicted ADHD liabilities. In addition, we summarized the results of the inverse-variance weighted method using a random-effects meta-analysis. Asthma prevalence was higher among participants with ADHD than those without ADHD (10.2% vs 5.9%). ADHD was associated with higher odds of asthma (odds ratio = 1.64, 95% confidence interval = 1.36–1.98), and the pattern of results was broadly consistent across subgroups. In contrast, the MR meta-analysis did not support a causal effect of genetically predicted ADHD liability on childhood asthma (odds ratio = 0.94, 95% confidence interval = 0.84–1.05, $P > .05$). According to data from the NHANES, ADHD and asthma are significantly associated in children and adolescents. A meta-analysis of the MR data did not reveal a causal connection between genetically predicted ADHD liability and childhood asthma. This paradoxical finding illustrates the complex interplay of genetic and nongenetic mechanisms in the ADHD-asthma association.

Abbreviations: ADHD = attention-deficit/hyperactivity disorder, BMI = body mass index, BMIZ = body mass index *Z*-score, CDC = Centers for Disease Control and Prevention, CI = confidence interval, ESR = estrogen receptor, G×E = gene-environment interactions, GWAS = genome-wide association study, HI = hyperactivity–impulsivity, IN = inattention, IVs = instrumental variables, IVW = inverse-variance weighted, MR = Mendelian randomization, NHANES = National Health and Nutrition Examination Survey, OR = odds ratio, PGC = Psychiatric Genomics Consortium, PIR = poverty-to-income ratio, SDs = standard deviations, SNPs = single-nucleotide polymorphisms, SP = study participant.

Keywords: ADHD, adolescents, asthma, children, Mendelian randomization, meta-analysis, NHANES

XL and JS contributed to this article equally.

This work was supported by the Guangxi Natural Science Foundation (grant no. 2024GXNSFAA010147), the National Natural Science Foundation of China (grant no. 82260953), Guangxi Peak Discipline: Traditional Chinese Medicine (Chinese Materia Medica; Document No. 2 [2026] of the Department of Education of Guangxi for Scientific Research), Guangxi Natural Science Foundation (grant no. 2023GXNSFAA026251), and Guangxi Higher Education Undergraduate Teaching Reform Project (2023JGB241).

The authors have no conflicts of interest to disclose.

Data sharing not applicable to this article as no datasets were generated or analyzed during the current study.

This research on humans didn't need ethical approval or participant consent, adhering to local and institutional standards. Additionally, our data came from public databases, eliminating the need for further ethical approval.

Supplemental Digital Content is available in the online version of this article (http://dx.doi.org/10.1097/MD.0000000000048878).

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How to cite this article: Lan X, Zhong N, Tang Z, Tang Z, Liu J, Leng X, Huang J, Sun J, Ai J. The association between asthma and ADHD in children and adolescents: An observational study and a meta-analysis of Mendelian randomization studies. Medicine 2026;105:21(e48878).

Received: 24 December 2025 / Received in final form: 26 February 2026 / Accepted: 29 April 2026

http://dx.doi.org/10.1097/MD.0000000000048878

1. Introduction

Attention-deficit/hyperactivity disorder (ADHD) is one of the most common neurodevelopmental disorders in childhood and is associated with substantial functional impairment and comorbid conditions. Asthma is a prevalent chronic respiratory disease in children and adolescents.[1–5] Increasing evidence suggests that ADHD and asthma may co-occur more often than expected.[6]

Prior observational studies have reported associations between asthma and ADHD, but findings have not been fully consistent across populations and study designs.[7–9] Proposed explanations include shared genetic susceptibility, environmental exposures, psychosocial factors, and inflammatory pathways.[10–13] Mendelian randomization (MR) has been used to assess potential causality between ADHD and asthma, yet published MR results remain mixed, likely due to differences in datasets and analytical approaches.[14–16] Two-sample MR leverages genetic variants as instrumental variables (IVs) to estimate the effect of an exposure on an outcome and can help reduce confounding and reverse causation.[17]

Therefore, we conducted a cross-sectional analysis using National Health and Nutrition Examination Survey (NHANES) to evaluate the association between ADHD and asthma in children and adolescents, and a meta-analysis of 5 two-sample MR studies to examine the effect of genetically predicted ADHD liability on childhood asthma risk. We hypothesized that ADHD would be associated with higher odds of asthma in NHANES and that genetically predicted ADHD liability would increase the risk of childhood asthma in MR.

2. Materials and methods

2.1. Observational epidemiological analysis

Three NHANES cycles (1999–2004) were selected for the study, as these cycles which data on diagnosed with ADHD in children and adolescents was available. Initially, 11,782 subjects aged 4 to 19 were identified in NHANES 1999 to 2004. Following the exclusion of 12 subjects with missing or indeterminate asthma status data were excluded, 24 subjects with incomplete ADHD data, and 1325 subjects with missing covariates (age, sex, race, poverty-to-income ratio [PIR], smoking status of household members, body mass index *Z*-score [BMIZ], total energy intake in 24 hours, and total fat intake in 24 hours), 10,421 subjects were deemed eligible for inclusion in this study's analysis. As illustrated in Figure 1, the screening process for participants is a multifaceted procedure.

2.2. Assessment of ADHD and asthma outcomes

In NHANES, asthma and ADHD were ascertained from the Medical Conditions Questionnaire (MCQ) module using self- or parent/guardian proxy-reported physician/health-professional diagnoses. The diagnosis of asthma was made by answering "yes" to the following question on the NHANES Family Questionnaire (MCQ: MCQ010): "Has a doctor or other healthcare professional told you/the study participant (SP) that you/SP has asthma?" To assess ADHD, in our study, ADHD was diagnosed by answering (MCQ: MCQ060) "Yes" to the following question: "Has a doctor or health professional ever told you/SP that you/he/she/SP has attention deficit disorder?"

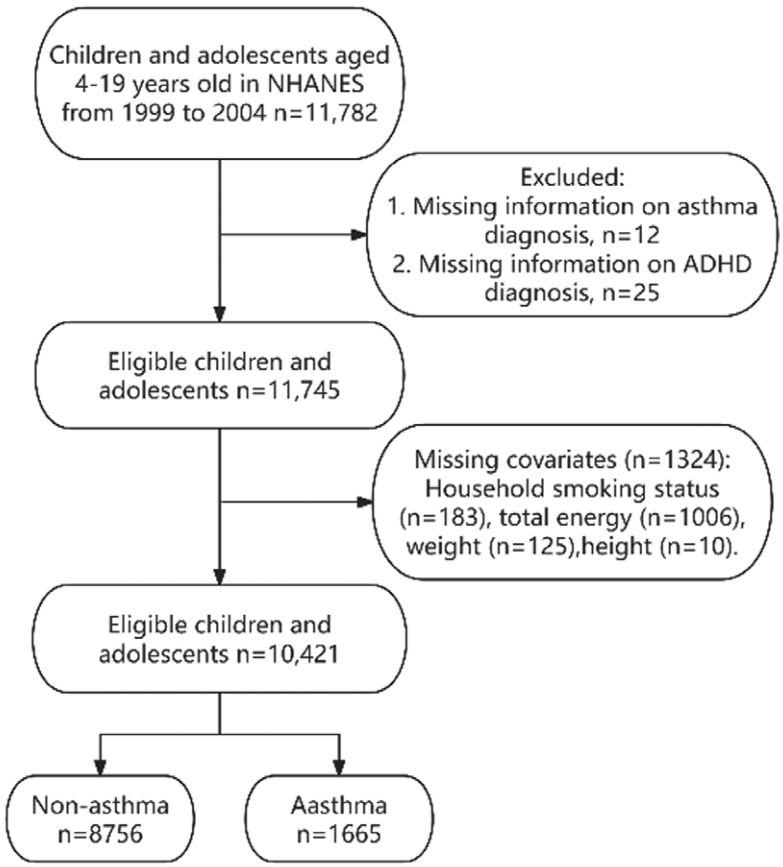


Figure 1. Study population selection flowchart. ADHD = attention-deficit/hyperactivity disorder, NHANES = National Health and Nutrition Examination Survey.

2.3. Other covariates

We identified potential confounders based on clinical insights and previous studies,[18–24] including age, sex, race, PIR, smoking status of household members, BMIZ, total energy intake over 24 hours, and total fat intake over 24 hours. PIR categorizes poverty status into 3 categories: 1.30, 1.31 to 3.50, and 3.50.

Body mass index (BMI) was calculated as weight in kilograms divided by height in meters squared (kg/m^2). Due to the lack of pre-calculated BMIZ data for the NHANES 1999 to 2004 cycles, BMI *Z*-scores were calculated based on age and sex using 2 distinct methods according to age groups:

For children aged ≤59 months, BMIZ was calculated using the WHO Anthro Survey Analyser Online (https://worldhealthorg.shinyapps.io/anthro/), following the tool's instructions and based on the 2006 WHO Child Growth Standards. Input parameters included age (in months), sex, standing height, and weight. BMIZ values outside the biologically plausible range of −5 to +5 standard deviations (SDs) were flagged as potential errors; in this study, no such extreme values were identified, and all calculated BMIZ values were retained for analysis.

For children and adolescents aged >59 months, BMIZ was calculated using SAS software (version 9.4; SAS Institute Inc.) and the Centers for Disease Control and Prevention (CDC)-provided SAS source code program (cdc-source-code.sas), which implements the CDC 2000 growth reference. This program utilizes age (in months or days), sex, height (in cm), and weight (in kg) to compute age- and sex-specific BMI *Z*-scores. Similarly, BMIZ values outside the range of −5 to +5 SDs were considered biologically implausible; no such extreme values were found in our dataset.

The age cutoff of 59 months (approximately 4 years 11 months) was strictly applied to determine the calculation method. Children aged 4 to 5 years are classified according to the 2006 WHO child growth standards,[24] while children and adolescents aged 5 to 19 years are classified according to the 2007 growth reference values.[25] Children's weight status was subsequently categorized as: normal weight (BMIZ < +1 SD), overweight (BMIZ ≥ +1 SD and <+2 SD), and obesity (BMIZ ≥ +2 SD).

Smoking status among household members was assessed using the NHANES household questionnaire (SMQFAM: SMD410). Dietary intake data for two 24-hour periods were used to determine participants' energy (kcal) and fat (g) intakes. An average of the total energy intake and total fat intake over the 2 recall periods was calculated; if only the first day's data were available, the first day's value was used instead.

Among covariates, PIR had a skewed distribution, median imputation of missing values was performed using Empower stats software (X&Y Solutions, Inc.). For all covariates except PIR, the individual was excluded if any covariate was missing. In the final analysis, the data of all 10,421 people were complete. Because PIR was right-skewed and served only as an adjustment covariate, we imputed missing PIR with the median (a robust measure for skewed distributions) to preserve sample size and statistical power.

2.4. Sources of two-sample MR

This two-sample MR study aimed to investigate the causal effect of ADHD on the risk of childhood asthma. Genome-wide association study (GWAS) data for ADHD were obtained from the Psychiatric Genomics Consortium (PGC).[26] A total of 225,534 people were included in the study, including 38,691 ADHD patients and 186,843 controls. Single-nucleotide polymorphisms (SNPs) associated with ADHD at genome-wide significant levels were selected, and this study used $P < 5 \times 10^{-8}$ as the screening threshold for exposure factors and selected independent SNPs with $r^2 < 0.01$ and clump distance ≤5000 kb to avoid the influence of linkage disequilibrium on the results. In addition, we assessed the strength of the exposure factor using the *F*-statistic value calculated as follows: $f = \beta^2/SD^2$, where β denotes the effect value of SNPs on exposure and SD denotes the standard deviation. SNPs with *F*-statistic values <10 were excluded to avoid bias due to weak IVs.[27] The final 26 SNPs associated with ADHD were retained as IVs for subsequent MR analyses. Table S1 presents further information regarding the exposure and outcome datasets. A comprehensive overview of the SNPs associated with ADHD can be found in Table S2. Childhood asthma GWAS summary statistics were drawn from 5 large, publicly available datasets (FinnGen, UK Biobank, and 3 published GWASs) to maximize statistical power and to assess robustness across independent resources, prioritizing predominantly European-ancestry samples to reduce population stratification.[28,29] Potential sample overlap between the PGC ADHD GWAS and each asthma GWAS dataset was evaluated based on reported cohort composition; no overlap was expected. Methodological work indicates that, if overlap were present in a two-sample MR with weak instruments, the resulting bias would tend to move estimates toward the confounded observational association, whereas complete non-overlap yields conservative bias toward the null.[30]

2.5. Statistical analysis

NHANES uses sampling weights, strata, and primary sampling units. In this study, regression analyses were conducted without applying NHANES sampling weights or survey design variables; therefore, estimates should be interpreted as associations within the analytic sample rather than population-representative estimates. In observational epidemiologic analyses, we grouped participants according to the presence or absence of asthma, and descriptive analyses were performed for all participants. We expressed baseline characteristics as mean ± SD for continuous variables and as percentages for categorical variables.

Stratified analyses were performed to assess the association between ADHD and asthma within predefined subgroups (sex, age, race, PIR, BMIZ, total energy intake, total fat intake, household smoking status; see Table 2). A significant interaction ($P < .05$) indicates that the strength of the association between ADHD and asthma differs across the levels of the stratification variable.

Statistical analysis using logistic regression was performed to assess the association between ADHD and asthma. Three models were constructed: Model 1 (unadjusted model), Model 2 (sex, age), and Model 3 (age, sex, race/ethnicity, PIR, smoking status of household members, BMIZ, total energy intake over 24 hours, and total fat intake over 24 hours). $P < .05$ was considered statistically significant. The strength of the relationship was determined by calculating odds ratios (ORs) and 95% confidence intervals (CIs).

In MR, genetic variation determines the causal relationship between an exposure factor and an outcome. To analyze the causal relationship between ADHD and asthma in children and adolescents, we adopted the following methods: inverse-variance weighted (IVW), MR Egger, weighted median, simple mode, weighted mode, and MR-PRESSO.[31] Cochran *Q* test was used to estimate heterogeneity. EmpowerStats software (www.empowerstats.com) and R (version 4.3.1; https://www.r-project.org) were used to analyze all data.

The IVW results from the 5 two-sample MR analyses were combined using a random-effects meta-analysis method. The I^2 statistic was then used to assess the heterogeneity of each outcome from the different data sources. I^2 values of <25%, between 25% and 75%, and >75% indicate low, moderate, and high heterogeneity, respectively. A random-effects model was

Table 1

Characteristics of the study participants grouped by asthma status.

Characteristics	Total (n = 10,421)	Non-asthma (n = 8756)	Asthma (n = 1665)	*P* value
Gender (N, %)				<.001
Female	5232 (50.2%)	4474 (51.1%)	754 (45.3%)	
Male	5189 (49.8%)	4283 (48.9%)	911 (54.7%)	
Age (yr, N, %)				.204
<6	957 (9.2%)	807 (9.2%)	150 (9.0%)	
6–11	2949 (28.3%)	2506 (28.6%)	443 (26.6%)	
≥12	6515 (62.5%)	5443 (62.2%)	1072 (64.4%)	
Race (N, %)				<.001
Mexican American	3530 (33.9%)	3139 (35.8%)	391 (23.5%)	
Other Hispanic	437 (4.2%)	337 (3.8%)	100 (6.0%)	
Non-Hispanic White	2766 (26.5%)	2341 (26.7%)	425 (25.5%)	
Non-Hispanic Black	3262 (31.3%)	2595 (29.6%)	667 (40.1%)	
Other Race including multiracial	426 (4.1%)	344 (3.9%)	82 (4.9%)	
PIR (N, %)				.454
<1.30	4263 (40.9%)	3574 (40.8%)	689 (41.4%)	
1.30–3.50	4232 (40.6%)	3577 (40.9%)	665 (39.3%)	
≥3.50	1926 (18.5%)	1605 (18.3%)	321 (19.3%)	
BMIZ (N, %)				<.001
<+1	6614 (63.5%)	5678 (64.8%)	936 (56.2%)	
+1 to +2	2930 (28.1%)	2371 (27.1%)	559 (33.6%)	
≥+2	877 (8.4%)	707 (8.1%)	170 (10.2%)	
Household smoking status (N, %)				.108
No	8099 (77.7%)	6830 (78.0%)	1269 (76.2%)	
Yes	2322 (22.3%)	1926 (22.0%)	396 (23.8%)	
ADHD (N, %)				<.001
No	9738 (86.9%)	8242 (94.1%)	1496 (89.8%)	
Yes	683 (6.6%)	514 (5.9%)	169 (10.2%)	
Total energy (kcal, mean ± SD)		2132.19 ± 931.69	2166.74 ± 954.79	.167
Total fat (g, mean ± SD)		78.10 ± 40.36	80.48 ± 41.88	.029

ADHD = attention-deficit/hyperactivity disorder, BMIZ = body mass index Z-score, PIR = poverty-to-income ratio, SD = standard deviation.

used when I^2 was >75%. Figure 2 shows the study design of the MR analysis and meta-analysis.

3. Results

3.1. Epidemiological observation and analysis

Finally, a total of 10,421 patients were included and divided into an asthma group and a non-asthma group, with 1665 patients (16.0%) in the asthma group and 8756 patients (84.0%) in the non-asthma group. There were significant differences between the asthma group and the non-asthma group in terms of gender, race, BMI *Z*-score, and ADHD prevalence ($P < .001$). In the asthma group, 54.7% were male, 40.1% were non-Hispanic Black, and the BMI *Z*-score distribution was skewed toward to higher intervals. Total fat intake was slightly higher in the asthma group than in the non-asthma group ($P = .029$). Notably, the ADHD prevalence rate was 10.2% in the asthma group, significantly higher than the 5.9% in the non-asthma group, suggesting a potential association between the 2 conditions (Table 1).

Stratified analyses (Table 2) generally showed a positive association between ADHD and asthma across demographic, socioeconomic, and metabolic strata. The magnitude of the association varied across subgroups (e.g., sex, age group, race/ethnicity, PIR, BMI category, dietary intake, and household smoking status), and interaction testing suggested effect modification by income (*P* for interaction < .05).

Table 3 shows the results of logistic regression analysis. ADHD was significantly and positively associated with asthma in all 3 models. The results showed that after adjusting for all covariates, the association between ADHD and asthma remained significant with an OR of 1.64 (95% CI = 1.36–1.98, *P* for trend < .01).

3.2. Results of MR analysis

The results of the MR analysis of ADHD on childhood asthma are shown in Table S3. Following our defined screening criteria ($P < 5 \times 10^8$, $r^2 < 0.01$, $F > 10$) and after excluding potential confounding factors associated with asthma, 26 independent SNPs significantly associated with ADHD were selected from the PGC dataset as genetic instrument variables for ADHD. The MR scatter plot (Fig. S1) visualizes the directional consistency of SNP effects on ADHD and childhood asthma. We conducted 5 two-sample MR analyses. The childhood asthma dataset from FinnGen Database included 8402 cases and 259,839 controls, all from a European population. The IVW method results showed a negative association between ADHD and childhood asthma (OR = 0.821, 95% CI = 0.695–0.969, $P < .05$), and the weighted median method also yielded a negative association (OR = 0.748, 95% CI = 0.623–0.898, $P < .05$). Funnel plot analysis did not reveal significant bias (Fig. S2A), and sensitivity analysis using the leave-one-out method indicated that removing any SNP did not significantly alter the results further supporting the robustness of the findings (Fig. S3A). The dataset from Sakaue et al's study[32] included 27,712 cases of European ancestry, 411,131 controls of European ancestry, 547 cases of East Asian ancestry, and 161,803 controls of East Asian ancestry. The IVW method results showed a negative association between ADHD and childhood asthma (OR = 0.800, 95% CI = 0.724–0.886, $P < .001$), and the weighted median method also yielded a significant negative association (OR = 0.845, 95% CI = 0.760–0.941, $P < .05$). We conducted levels of multi-effect tests in both MR analyses, and the results indicated no significant level of multi-effect, with *P* values of .621 and .787, respectively. However, Cochran *Q* test results indicate significant heterogeneity ($P < .05$).

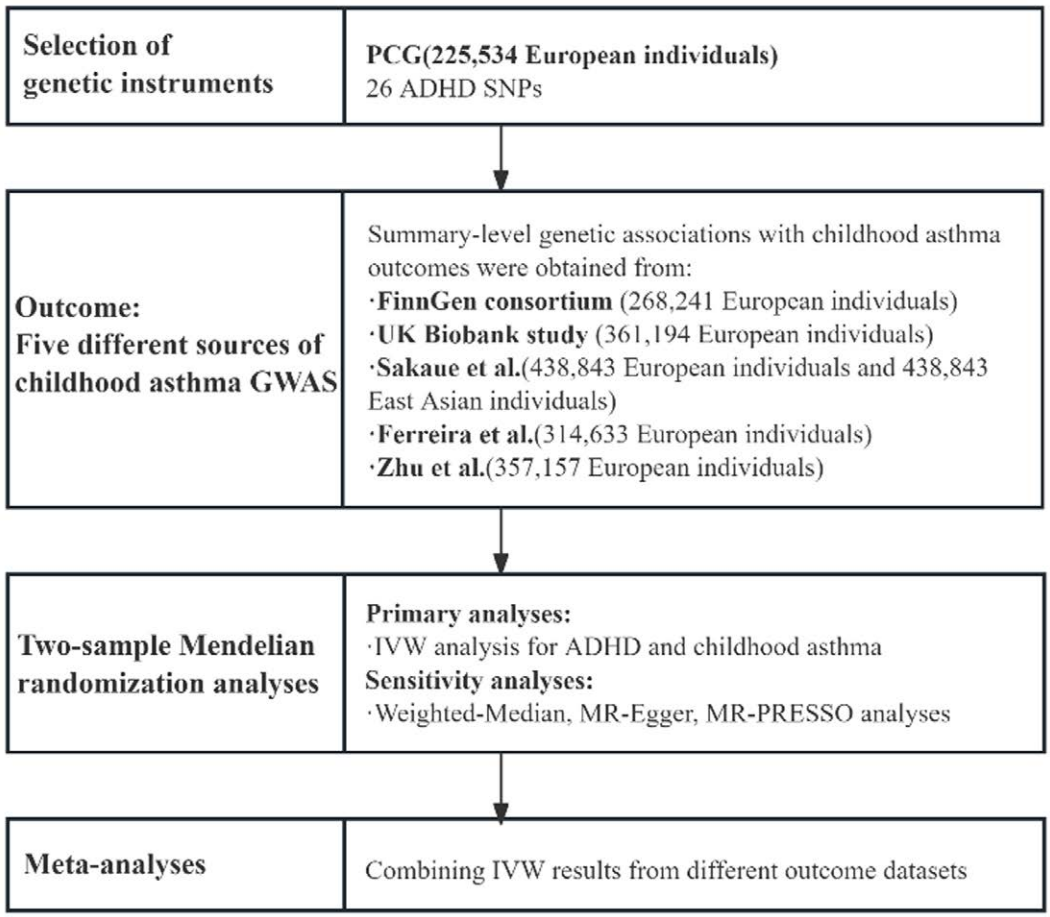


Figure 2. Design of MR analysis and meta-analysis studies for ADHD and childhood asthma. ADHD = attention-deficit/hyperactivity disorder, GWAS = genome-wide association study, IVW = inverse-variance weighted, MR = Mendelian randomization, PGC = Psychiatric Genomics Consortium, SNP = single-nucleotide polymorphisms.

The results from the Ferreira et al,[33] Zhu et al,[15] and UK Biobank datasets showed no significant causal association between ADHD and childhood asthma. In particular, the dataset from the Ferreira et al study included 13,962 cases and 300,671 controls with a European population, and the results of the IVW method showed no significant association (OR = 1.031, 95% CI = 0.932–1.140, $P = .552$), and the weighted median method suggested no significant effect (OR = 1.023, 95% CI = 0.908–1.152, $P = .714$). The dataset from Zhu et al included 9676 cases and 347,481 controls, and the population was European. Both the IVW method (OR = 1.064, 95% CI = 0.906–1.249, $P = .449$) and the weighted median method (OR = 1.084, 95% CI = 0.886–1.326, $P = .421$) results suggested no significant causal relationship. The UK Biobank dataset included 1993 cases and 359,201 controls, and the IVW method (OR = 0.999, 95% CI = 0.997–1.000, $P = .093$) and weighted median method (OR = 1.000, 95% CI = 0.998–1.002, $P = .804$) results similarly suggested no significant causal relationship between ADHD and childhood asthma. We tested for horizontal pleiotropy in all 3 of the above MR analyses, and the results were obtained with P values of .583, .298, and .680, suggesting no significant horizontal pleiotropy (Table S3 and Fig. S3C–E).

3.3. Meta-analysis results

This meta-analysis identified significant heterogeneity ($I^2 = 83.9\%$, $P < .0001$) across studies, necessitating a random-effects IVW model. The pooled OR (OR = 0.94, 95% CI = 0.84–1.05, $P > .05$) across FinnGen, UK Biobank, and prior studies (Sakaue et al, Ferreira et al, and Zhu et al) demonstrated no significant causal relationship between ADHD and childhood asthma risk (Fig. 3).

4. Discussion

This study examined the association between ADHD and asthma in children and adolescents using observational analyses and MR. Observational analyses indicated an association between ADHD and asthma, whereas MR did not support a causal effect. This apparent discrepancy highlights the complex interplay of genetic and nongenetic mechanisms underlying ADHD–asthma comorbidity. Because the NHANES component is cross-sectional, these observational findings should be interpreted as associations and do not establish causation. In contrast, MR evaluates the effect of genetically predicted ADHD liability on asthma under instrumental-variable assumptions.

We observed a positive association between ADHD and asthma, consistent with previous studies.[34,35] For example, a case–control study in adults reported a significant association: asthma prevalence was 9.7% among individuals with ADHD, significantly higher than in those without ADHD.[34] In addition, a meta-analysis suggested that children with ADHD are more likely to develop allergic diseases, including asthma.[35]

Stratified analyses indicated that the ADHD–asthma association was stronger in females, suggesting that sex may modify mechanisms underlying comorbidity. For ADHD, pediatric clinical studies report a male-to-female diagnostic ratio of ~4:1,[36] whereas population-based studies often report a ratio closer to 2:1.[37] Females with ADHD more often present with

Table 2

Subgroup analyses for the association of ADHD and asthma.

Parameters	N	OR (95% CI)	*P* value	*P* for interaction
Sex				.204
Female	5232	2.09 (1.49–2.94)	<.0001	
Male	5189	1.60 (1.29–2.00)	<.0001	
Age				.437
<6	957	2.43 (0.74–7.99)	.1441	
6–11	2949	2.13 (1.52–2.99)	<.0001	
≥12	6515	1.68 (1.34–2.09)	<.0001	
Race				.143
Mexican American	3530	2.24 (1.41–3.56)	.0006	
Non-Hispanic Black	3262	1.56 (1.15–2.13)	.0046	
Non-Hispanic White	2766	1.60 (1.18–2.16)	.0022	
Other Hispanic	437	3.79 (1.78–8.06)	.0005	
Other Race including multiracial	426	1.10 (0.43–2.80)	.8385	
PIR				.031
<1.30	4263	2.36 (1.78–3.12)	<.0001	
1.31–3.49	4232	1.53 (1.13–2.08)	.0056	
≥3.50	1926	1.47 (0.98–2.19)	.0616	
BMIZ				.904
<+1	6614	1.80 (1.41–2.28)	<.0001	
+1 to +1.9	2930	1.78 (1.29–2.46)	.0004	
≥+2	877	2.07 (1.13–3.78)	.0179	
Total energy				.761
Low	3474	1.73 (1.19–2.51)	.0044	
Middle	3473	1.65 (1.17–2.33)	.0047	
High	3474	1.92 (1.48–2.51)	<.0001	
Total fat				.719
Low	3474	1.96 (1.37–2.82)	.0003	
Middle	3473	1.89 (1.34–2.65)	.0003	
High	3474	1.66 (1.26–2.17)	.0003	
Household smoking status				.911
No	8099	1.81 (1.45–2.26)	<.0001	
Yes	2322	1.77 (1.28–2.45)	.0006	

ADHD = attention-deficit/hyperactivity disorder, BMIZ = body mass index Z-score, CI = confidence interval, OR = odds ratio, PIR = poverty-to-income ratio.

Table 3

Associations of ADHD and asthma in US children and adolescents.

	Mode 1		Mode 2		Mode 3	
Expose	**OR (95% CI)**	***P***	**OR (95% CI)**	***P***	**OR (95% CI)**	***P***
ADHD	1.81 (1.51–2.17)	<.001	1.73 (1.44–2.08)	<.001	1.64 (1.36–1.98)	<.001

ADHD = attention-deficit/hyperactivity disorder, CI = confidence interval, OR = odds ratio.

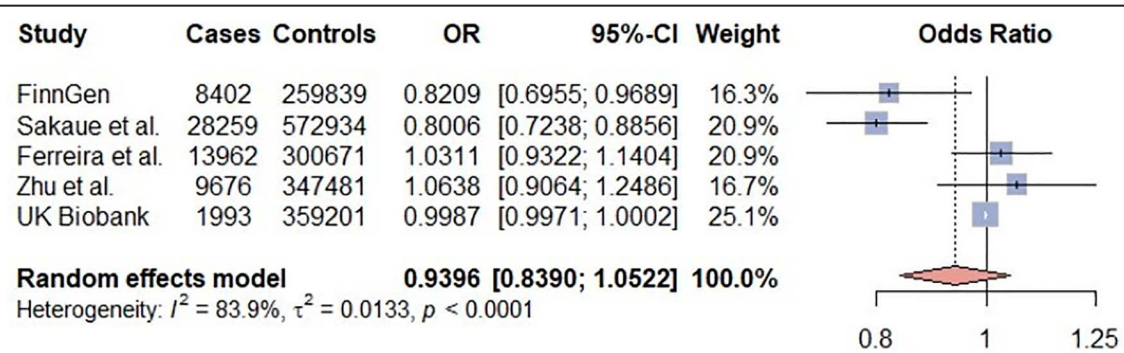


Figure 3. Forest plot of genetic causality between ADHD and childhood asthma outcomes assessed by IVW MR. *P* = *P* value for causal estimation. Heterogeneity interpretation (I^2): <25% (low), 25% to 75% (moderate), >75% (high). ADHD = attention-deficit/hyperactivity disorder, CI = confidence interval, IVW = inverse-variance weighted, MR = Mendelian randomization, OR = odds ratio.

internalizing symptoms (e.g., inattention [IN]), and are frequently diagnosed later than males.[38] For asthma, a Swedish national cohort study reported that asthma incidence in females exceeds that in males after puberty,[39] suggesting that pubertal hormonal changes may influence asthma pathogenesis. In addition, a GWAS in Han Chinese children identified a significant association between estrogen receptor (ESR) polymorphisms and ADHD susceptibility,[40] the ESR1 variant rs6930355

showed a trend towards association in females. ESR signaling has been implicated in sex differences across psychiatric disorders, suggesting that hormonal factors may modulate ADHD pathophysiology. Notably, the same variant (rs6930355) has also been associated with obesity-related traits (e.g., waist circumference and BMI),[41] highlighting its broader involvement in metabolic and neurodevelopmental pathways. Animal studies provide further insight into sex-specific mechanisms. Male ESR1-knockout mice showed reduced airway hyperresponsiveness after allergen exposure ($P < .05$), whereas no such effect was observed in females.[42] This finding suggests that ESR1 regulation of airway hyperresponsiveness differs by sex, with males exhibiting a more pronounced airway response to allergens. Further research is needed to determine whether hormone fluctuations across developmental stages influence the co-occurrence of ADHD and asthma.

Age-stratified analyses showed a significant association between ADHD and asthma among participants aged 6 to 19 years, consistent with prior epidemiological evidence.[43] Intriguingly, the <6 years age group (N = 957) showed the highest point estimate of risk (OR = 2.43, 95% CI = 0.74–7.99, $P = .144$), though statistical significance was not achieved, potentially due to limited power from the small subsample size. However, children of different age groups have different immune levels[44] and sex hormone levels, especially during puberty, which is an important developmental period marked by hormonal, metabolic, and immune changes,[45] with hormone-driven gender differences gradually becoming apparent.[46,47] In the future, it will be important to conduct further longitudinal studies.

Race/ethnicity-stratified analyses indicated a stronger ADHD–asthma association among Hispanic participants, particularly in other Hispanic subgroups. Differences in educational resources, socioeconomic gradients, parental stress, and cumulative adverse childhood experiences exist across racial groups,[48–52] which may simultaneously increase the likelihood of ADHD and asthma in certain racial/ethnic groups. We also observed that the association was strongest in low-income households. Studies have reported that each 10% increase in household income during childhood is associated with a 0.6% to 1.7% increase in asthma medication use and a 2.5% decrease in psychotropic medication use.[53] Economic stress may act synergistically through exacerbating psychosocial stressors (such as strained caregiver-child relationships[54]) and environmental triggers (such as homelessness[55]), thereby increasing the risk of neurodevelopmental disorders and allergic diseases.

Stratified analyses by BMI *Z*-score (BMIZ), total energy intake and dietary fat intake indicated stronger associations among overweight/obese participants and those with high-calorie, high-fat diets. Although ADHD co-occurs with obesity,[56] and obesity is associated with asthma onset,[57,58] ADHD and overweight/obesity in the general population may also reflect psychosocial factors (e.g., overcrowded living conditions and parental separation).[59] High energy intake may influence both ADHD and asthma.[60,61] For example, high consumption of artificially sweetened beverages has been linked to hyperactive symptoms.[62] In addition, early use of stimulant medications (e.g., methylphenidate) may increase susceptibility to sugar reinforcement, thereby increasing the risk of obesity.[63] Notably, interactions along the gut–brain and gut–lung axes have been proposed as possible pathways linking neurodevelopmental and respiratory phenotypes, primarily through immune modulation and microbiota-related processes.[64–69] Both ADHD and asthma have been reported to be associated with low-grade inflammatory activity, and cytokines such as interleukin-6 and tumor necrosis factor-α have been discussed as candidate correlates rather than confirmed drivers of co-occurrence.[70] High-sugar/high-fat dietary patterns can alter gut microbiota composition and intestinal barrier function and may influence systemic inflammation.[71] In children with ADHD, differences in gut microbial profiles and related metabolites (including short-chain fatty acids) have been observed, and SCFA-related immune effects have been suggested to be relevant to allergic disease and asthma.[72–75] However, these gut–brain/lung pathways remain hypothesis-generating, and longitudinal or experimental studies are needed to establish directionality and causality.

The association between ADHD and asthma was significant regardless of the presence of smokers in the home. Previously, exposure to secondhand smoke in the home has been investigated as a potential risk factor for ADHD associated with allergic diseases.[76] Harmful substances from smoking (e.g., nicotine and tar) may increase the risk of ADHD and asthma through multiple mechanisms, such as affecting fetal development and influencing the function of the immune system in children.[77,78]

The results of the stratified analyses in this study suggest that factors such as gender, age, race, household economic status, weight, and dietary habits may influence the association between ADHD and asthma through different mechanisms. These findings provide important directions for future research, suggesting the need to further explore the specific roles of these factors in the mechanisms of ADHD and asthma. However, observational studies are vulnerable to confounding, including unmeasured factors such as maternal medication use during pregnancy and maternal disease history. Genetic studies have found that parental asthma, particularly maternal asthma exacerbation, is associated with an increased risk of ADHD in offspring.[79] However, there is limited evidence and conflicting results regarding a causal relationship between ADHD and asthma. In order to assess the causal relationship, we used MR and meta-analysis of MR results.

The results of 5 two-sample MR analyses were inconsistent, and this discrepancy may be attributed to differences in population characteristics and sample sizes across the datasets.[80,81] ADHD in childhood is highly heritable and primarily driven by familial factors; however, its heritability appears to be lower in adulthood, though this remains a topic of ongoing debate.[11] A study in Turkey observed that the prevalence of asthma among children with ADHD was similar to that of their siblings.[82] Some studies have separated ADHD into hyperactivity–impulsivity (HI) and IN dimensions and examined each in relation to asthma.[83] They found that the association between childhood asthma and HI may be mediated by genetic overlap or environmental factors, while no significant association was found for IN.[84] Future studies could refine phenotyping by analyzing HI and IN separately and by classifying asthma into clinically relevant subtypes (e.g., cough-variant asthma), to test for subtype-specific causal relationships.

Although MR can reduce unmeasured confounding and reverse causality, ADHD and asthma are influenced by both genetic and environmental factors,[84,85] and MR primarily captures effects mediated by genetic instruments. Secondly, limitations of IVs are also a possible reason for this discrepancy, and despite the stringent screening criteria we used ($P < 5 \times 10^{-8}$, $r^2 < 0.01$), the IVs (SNPs) may still suffer from pleiotropy,[86] which affects the accuracy of causal inference. The present study included a dataset sample of mainly children of European origin, which may have reached different conclusions from previous observational studies due to differences in genetic background. Meta-analysis suggested a high degree of heterogeneity between studies ($I^2 = 83.9\%$), suggesting that different populations, environments, or genetic backgrounds may have contributed to the differences in effect sizes.

4.1. Significance and limitations of the study

Our study integrates data from multiple annual cycles and specifically targets children and adolescents aged from 4 to 19 years. MR studies reduce confounding bias and reverse causality

bias. We also performed a meta-analysis to combine associations estimated from independent data sources. Providing multidimensional evidence for comorbid ADHD and asthma, this approach ensured adequate statistical power and robustness of the findings.

As the main study population was children and adolescents aged 4 to 19 years, some covariates (e.g., smoking status, alcohol consumption, mental health problems) were incomplete, thus preventing correlation analyses and confirmation of whether these covariates played an interaction role in this study. Our assessment of ADHD and asthma relied solely on parent/guardian self-report in the NHANES family questionnaires. Although this approach is practical for large-scale surveys, it is susceptible to misclassification. Self-reported diagnoses may underestimate the true prevalence, particularly for milder cases that do not seek medical attention or where the diagnosis was not communicated or recalled.[87,88] Furthermore, differential reporting tendencies or healthcare access across racial/ethnic groups and socioeconomic strata could introduce bias in the observed associations.[89,90] This non-differential misclassification would likely attenuate the observed associations towards the null, meaning our reported ORs might underestimate the true strength of the link between ADHD and asthma.

While MR offers advantages over observational studies in minimizing confounding and reverse causality, several inherent limitations of the MR approach warrant careful consideration when interpreting our causal null findings:

MR primarily captures the component of the causal effect mediated by the genetic predisposition to the exposure (ADHD). Both ADHD and asthma are complex multifactorial disorders influenced by intricate interplay between genetic susceptibility and environmental exposures (e.g., air pollution, early-life infections, psychosocial stress, and diet). Our analysis cannot incorporate direct effects of environmental factors or gene-environment interactions (G × E). For instance, the genetic risk for ADHD might only translate to increased asthma risk under specific environmental triggers, a pathway invisible to standard MR.

Asthma is a highly heterogeneous condition. Our MR analyses primarily utilized broadly defined "childhood asthma." Crucially, ADHD genetic liability might be causally linked only to specific asthma endotypes or phenotypes, such as allergic asthma or severe asthma, rather than asthma as a whole. The aggregation of diverse subtypes in the outcome GWAS could dilute or obscure genuine subtype-specific causal effects. Future MR using GWAS data on well-defined asthma subtypes is needed.

Therefore, our MR meta-analysis, while not supporting a consistent direct causal effect of the genetically proxied risk for (largely adult-defined) ADHD on broadly defined childhood asthma, cannot rule out: environmental effects or G × E underlying the comorbidity; a causal effect specific to childhood ADHD genetic factors or specific asthma subtypes; or important indirect causal effects mediated by ADHD-driven behavioral and lifestyle changes.

5. Future research directions

In the future, multi-omics analyses combining genomic, epigenomic, and transcriptomic data should be performed to reveal the molecular mechanisms underlying the co-morbidity between ADHD and asthma. Longitudinal studies should be designed and prospective cohort studies should be conducted to assess the long-term association and dynamic changes of ADHD and asthma. Asthma is a heterogeneous disease, and different subtypes (e.g., allergic asthma vs nonallergic asthma) may be differently associated with ADHD. In addition to the traditional classifications of ADHD into hyperactive-impulsive, inattentive, and combined (ADHD-C) subtypes, ADHD can also be categorized into familial and nonfamilial forms.[91] These subtype classifications exhibit distinct developmental trajectories and clinical outcomes.[92] Large-scale stratified analyses are crucial for resolving subtype-specific heterogeneity and identifying potential causal pathways.

6. Conclusion

Using observational and MR analyses, we found a positive ADHD–asthma association in NHANES but no causal effect in MR, with heterogeneity across asthma GWAS datasets. These results suggest complex genetic and nongenetic contributions. Clinicians should consider integrated management for children with co-occurring ADHD and asthma, and coordinated pediatric care may improve outcomes and reduce healthcare utilization.

Acknowledgments

We would like to thank all participants in this study.

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